

Part I: Fitting diffusion tensors from MRI imagery

Introduction

Diffusion tensor imagery (DTI) is a usage for magnetic resonance imagery (MRI) used to gather more detail about the brain, which allows for visualisation of neural pathways and connectivity (Ranzenberger, et. al, 2023). However, extra data processing is required to visualise diffusion tensor imagery - namely, tensor calculation. An article by Hangyi Jiang, et. al, goes over the necessary theory and calculations required to perform computation of the DTI, and how this data can be used to create “*apparent diffusion constant maps, anisotropy maps, principal eigenvalue maps, eigenvector images, [and] color maps*” (Jiang, et. al, 2006). This report will go over our implementation of these calculations in MATLAB given raw DTI data. This will be accomplished using tensor fields, Gaussian radial basis functions, and eigendirection fields.

Background Theory

The theory behind DTI involves measuring the diffusivity of water within the brain tissue. A signal S_0 is recorded as a baseline, before a series of diffusion sensitising gradient pulses in the directions $\mathbf{g}$ are recorded. A value b is generated from other properties of the gradient pulses, such as its strength and timing. For each of the gradient directions $\mathbf{g}$, a signal S is also recorded such that using the equation

$$S = S_0 \exp(-b\mathbf{g}^T \mathbf{D} \mathbf{g})$$

(Queensland University of Technology, u.d.)

the diffusion tensor $\mathbf{D}$ can be calculated.

Methodology

Mathematical Model

For this use case, the diffusion tensor $\mathbf{D}$ is modelled by a 3x3 symmetric matrix

$$\mathbf{D} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{xy} & D_{yy} & D_{yz} \\ D_{xz} & D_{yz} & D_{zz} \end{bmatrix}$$

(Jiang, et. al, 2006)

A value for the six scalar elements of $\mathbf{D}$ can be estimated by assuming the equation

$$\log \left(\frac{S}{S_0} \right) = -b\mathbf{g}^T \mathbf{D} \mathbf{g}$$

(Queensland University of Technology, u.d.)

holds for each direction $\mathbf{g}$ in each voxel and solving using least squares.

MATLAB Implementation

Conclusion

Results

Discussion

Limitations

Improvements

Bibliography

Ranzenberger, L. R., Das, J. M., Snyder, T. (2023) *Diffusion Tensor Imaging*. National Institute of Medicine. [article] Available at: <https://www.ncbi.nlm.nih.gov/books/NBK537361/>
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